

Dimension	Option	What it is	What it implies
Attention	cross bidirectional (default)	The causal mask is passed as tgt_mask only, so self-attention is causal but cross-attention sees the whole history.	A history token can attend forward in time, so per-position states are not causal; fine for one query at the end of the sequence.
Attention	fully causal (– use_causal_memory)	The mask is passed as memory_mask too, so both blocks are triangular.	Every token uses its own past only, which is what a bedside model has; costs a little accuracy.
History	target analyte only (default)	Tokens are the patient’s previous draws of the analyte being predicted.	A clean univariate setpoint model; no other analyte can leak into the query.
History	full panel (– use_full_panel)	Every past draw across analytes, capped at 128, with a flag for whether the target analyte was drawn at that timestamp.	Multivariate and irregularly timed, so a creatinine trend can inform urea; longer, sparser sequences.
Covariates	sex, analyte, age at first draw (always on)	Context token carrying the demographics and assay identity.	Matches how the population interval is stratified, so a width difference is not a stratification difference.
Covariates	age at each draw (– use_age_t)	Binned age embedding on every history token, not just the context.	Lets the setpoint drift with age; matters over decades, not within an admission.
Covariates	care setting (– use_setting)	Inpatient / outpatient / ED embedding on each history token.	Separates an abnormal value from a value drawn in an abnormal place.
Covariates	same-draw co-analytes (– use_coanalytes)	The other analytes drawn at the same timestamp, as [value, observed, low, normal, high].	Adds the clinical context of each past draw; cannot see the query draw’s panel.
Covariates	co-analytes on the query too (–query_coanalytes)	The same panel encoding attached to the query token.	Conditions the interval on concurrent results, so it can no longer be used to flag them.
Split	sequence-level (default)	Splits are over patient-analyte sequences.	One patient’s potassium can train while their sodium tests: optimistic if any patient-level signal is learnable.
Split	patient-level (–split_by patient)	Every sequence from a patient falls in one split.	The conservative generalisation claim; not comparable with sequence-split arms, which score different rows.
Normalisation	within-sequence	Each history is standardised by its own mean and SD and the head’s output de-normalised.	One model can serve every analyte, and the head predicts deviation from the patient’s own level; a one-draw history has no usable SD.
Head	quantile (default)	Linear head emitting the five levels (2.5, 25, 50, 75, 97.5).	Distribution-free, but nothing enforces monotonicity or stops the interval collapsing on a short history.
Head	gate	Own quantiles plus a scalar gate g; returns $g * \text{own} + (1 - g) * \text{population quantiles}$.	A readable trust dial, bounded between the two inputs.
Head	nig	Conjugate normal-inverse-gamma head; the interval is the closed-form Student-t posterior predictive against the population prior.	Shrinkage is analytic, so one draw provably gives the population interval; assumes Gaussian within-person variation.
Head	gaussian	Mean and log-variance heads; interval is $\mu \pm 1.96 \text{ sigma}$.	Comparable likelihood with the Gaussian baselines; forces symmetric intervals.
Loss	QuantileLoss (pinball)	Mean pinball loss over the five levels.	Levels are fit independently; with few draws it shrinks toward the median, because that minimises pinball in-sample.
Loss	QuantilePriorLoss, anchor	Pinball plus the expected pinball under the population interval, weighted $k/(n+k)$, on normal-state queries only.	Bayesian shrinkage as a loss: no history gives the population interval, and the pull decays with draws.
Loss	QuantilePriorLoss, tau	The same with n time-decayed, $\sum(\exp(-dt/\tau))$.	Old draws stop counting, so a stale setpoint is not treated as well estimated.
Loss	QuantilePriorLoss, floor	Pinball plus a hinge when the predicted width falls below the population width.	Fixes over-narrow intervals without pinning location; can only widen, never centre.
Loss	QuantilePriorLoss, gate	Pinball on the gated quantiles plus a KL pulling the gate to $1 - k/(n+k)$.	The prior acts on self-trust rather than location, so shrinkage is inspectable per prediction.
Loss	StudentTNLLLoss	NLL of the NIG head’s Student-t predictive.	The only arm whose reported interval and training objective are the same object.
Loss	NORMALLoss (KL-aligned)	Gaussian NLL plus $\lambda * \text{KL}$ to the population distribution, weighted $k/(n+k)$ on normal queries and reversed on abnormal ones.	The original formulation; the abnormal-side push is a hinge, so its scale is arbitrary relative to the KL.
Prior	state-conditional population table	Fixed lookup [analyte, state, sex] of quantiles, μ , σ and ρ (within-person share of variance); never updated by gradient.	What every arm shrinks toward is frozen and auditable; ρ sets how much one draw is worth.
Prior strength	$k = 5$ vs $k = 20$	Pseudo-draws the prior is worth, in weight $k/(n+k)$.	$k = 5$ hands control to the patient after a few draws; $k = 20$ keeps most histories near population width.

Table 1: The design dimensions that distinguish the NORMA arms in Table ??: each row is one option along one dimension, what the model does and what the choice costs. Options marked (default) are what an arm uses when the flag is absent. Heads, losses and priors combine freely; the arms table records which combinations were trained.